

MAT 243 Project One Summary Report

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1. Introduction: Problem Statement

In this project I am working as a data analyst for the Golden State Warriors. The goal is to look at historical NBA game data and find useful patterns that can help the team's management make better decisions about improving performance. The data set I am using is the FiveThirtyEight NBA Elo dataset which contains historical game results including points scored, team names, game location, and a measure of relative team skill called the Elo rating. To analyze this data, I used three main statistical methods. First, I created data visualizations like histograms and boxplots to study how points were distributed across games. Second, I calculated descriptive statistics including the mean, median, variance, and standard deviation to understand the center and spread of the data. Third I constructed 95% confidence intervals to estimate the average relative skill level of all NBA teams during the years being studied. Together these methods give a well-rounded picture of team performance that can be easily communicated to management.

2. Introduction: Your Team and the Assigned Team

For this project I chose the Golden State Warriors with data from the years 2013 to 2015. The Warriors were assigned as my team to analyze their scoring patterns and overall performance during that time. The Chicago Bulls from 1996 to 1998 were assigned to me for a comparative study. The table below summarizes both teams and their respective years.

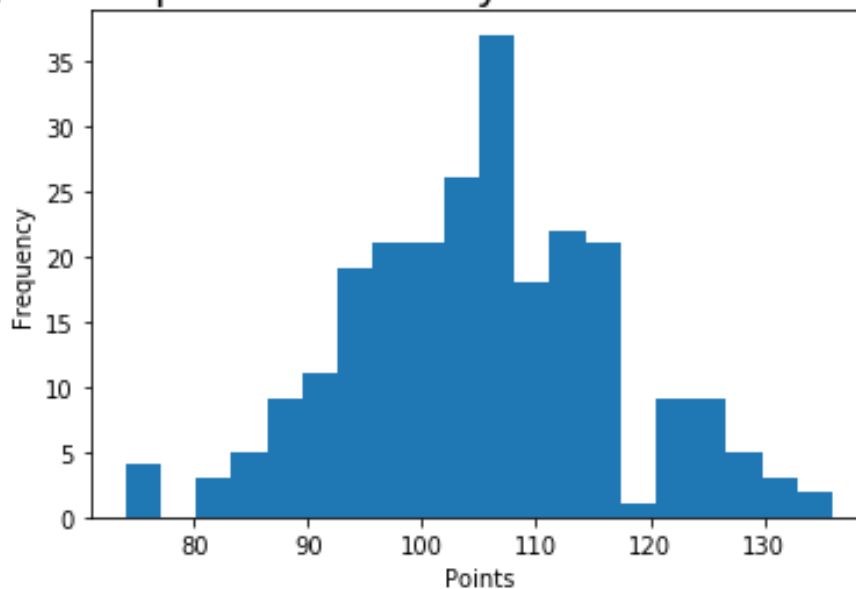
Table 1. Information on the Teams

	Name of Team	Assigned Years
1. Yours	Warriors	2013 - 2015
2. Assigned	Bulls	1996 - 1998

3. Data Visualization: Points Scored by Your Team

Data visualization is a useful tool for studying how data is distributed and spotting trends that might not be obvious just by looking at numbers. For the Golden State Warriors, I chose the histogram to display the distribution of points scored from 2013 to 2015. I picked the histogram over the scatterplot because it does a better job of showing how often certain point totals occurred across all games which makes it easier to see the overall shape of the data. Looking at the histogram, the Warriors scored anywhere between 80 and 135 points per game with most games landing somewhere between 100 and 115 points. The distribution appears roughly bell shaped which means extreme scoring games on either end were not very common. This tells us that the Warriors were a consistent scoring team during this time and that their performance did not fluctuate too dramatically from game to game.

Histogram of points scored by Your Team in 2013 to 2015

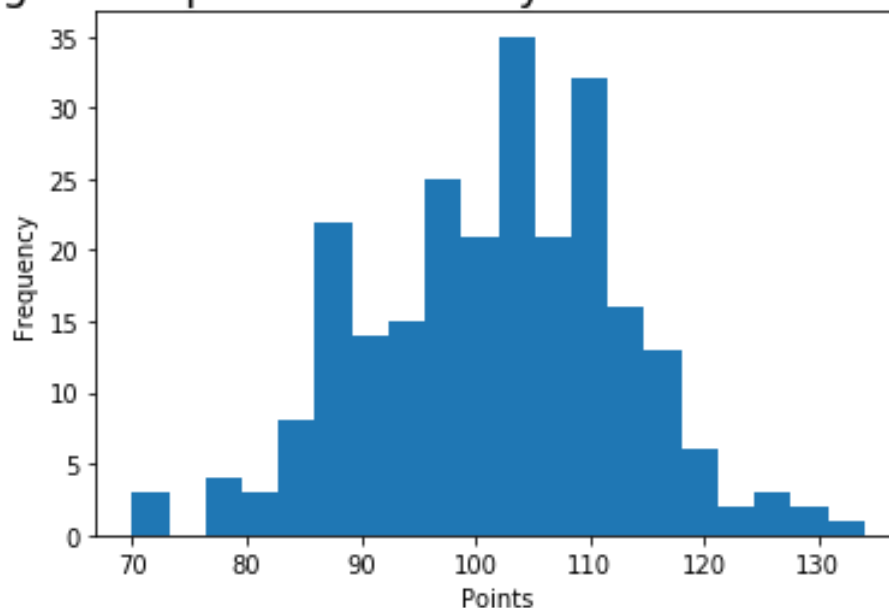


4. Data Visualization: Points Scored by the Assigned Team

For the Chicago Bulls I also chose the histogram to display the distribution of points scored from 1996 to 1998. I picked the histogram over the scatterplot for the same reason as before because it

gives a clearer picture of how often certain point totals occurred and makes it easier to see the overall shape of the distribution. Looking at the Bulls histogram, the team scored anywhere between 70 and 135 points per game with most games falling between 90 and 110 points. The distribution also appears roughly bell shaped, which means the Bulls were also a consistent scoring team during their dominant years in the late 1990s. However, compared to the Warriors the Bulls scoring range was slightly wider and their scores were centered a little lower overall. This suggests that while the Bulls were competitive and consistent they did not score as frequently in the higher point ranges as the Warriors did during their respective time periods.

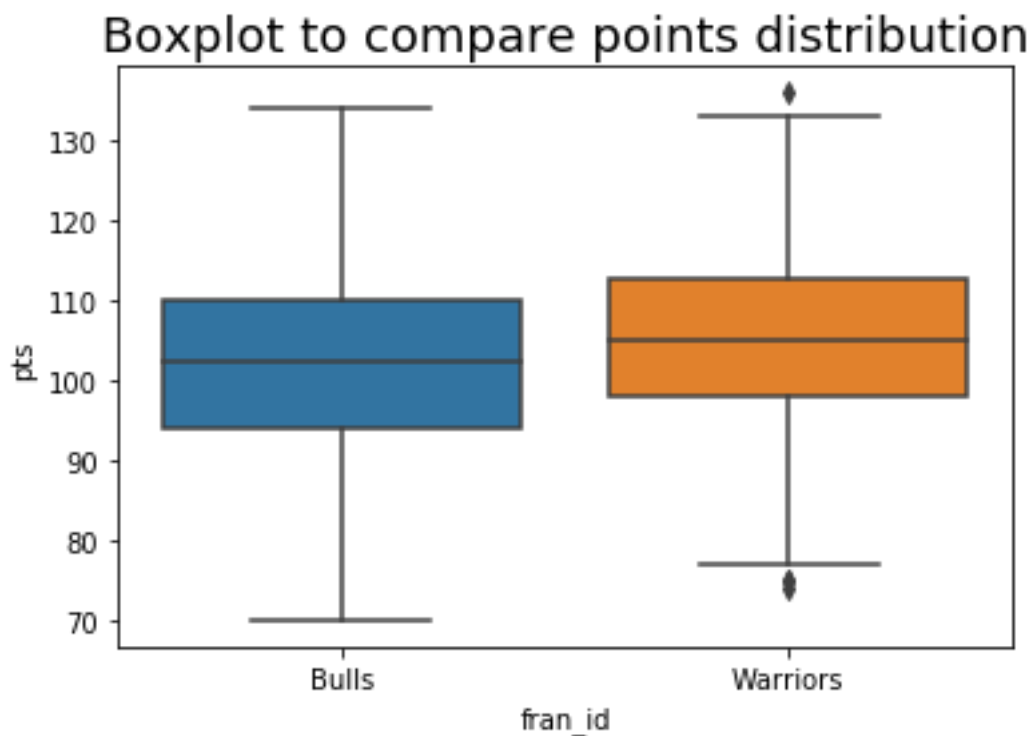
Histogram of points scored by the Bulls in 1996 to 1998



5. Data Visualization: Comparing the Two Teams

Data visualization is a great way to compare two different data distributions because it lets you see the differences and similarities between them quickly without having to dig through a bunch of numbers. For comparing the Golden State Warriors and the Chicago Bulls I chose the boxplot because it does the best job of showing the differences between the two teams side by side. The

boxplot clearly displays the median, the spread of the data, and any outliers all in one simple visual which makes it easier to compare the two teams at a glance. Looking at the boxplot, the Warriors had a higher median score than the Bulls overall which means the Warriors tended to score more points per game during their respective time. The Bulls had a slightly wider spread in their data which tells us their scoring was a little more unpredictable from game to game. Overall, the Warriors appear to have been the more consistent and higher scoring team when comparing the two distributions visually.



6. Descriptive Statistics: Points Scored by Your Team in Home Games

The measures of central tendency like the mean and median and measures of variability like the variance and standard deviation are used to summarize and understand how data is distributed. The mean tells us the average value while the median tells us the middle

value. The variance and standard deviation tell us how spread out the data is from the average. Together these statistics give us a well-rounded picture of the data distribution.

For the Warriors home games from 2013 to 2015 the descriptive statistics are summarized in the table below.

Table 2. Descriptive Statistics for Points Scored by Your Team in Home Games

Statistic	Value
Mean	106.62
Median	106.00
Variance	136.76
Standard Deviation	11.69

The Warriors averaged 106.62 points per game at home with a median of 106 points. Since the mean and median are very close together the distribution is roughly bell shaped meaning it is not heavily skewed in either direction. Because the distribution is roughly symmetric the mean is the best measure of central tendency to represent the center of this distribution. The standard deviation of 11.69 tells us that most games fell within about 11 to 12 points of the average which shows the Warriors were a consistent team when playing at home.

7. Descriptive Statistics: Points Scored by Your Team in Away Games

For the Warriors away games from 2013 to 2015 the descriptive statistics are summarized in the table below.

Table 3. Descriptive Statistics for Points Scored by Your Team in Away Games

Statistic Name	Value
Mean	103.63
Median	104.00
Variance	130.77
Standard Deviation	11.44

The Warriors averaged 103.63 points per game in away games with a median of 104 points.

Since the mean and median are very close together the distribution is roughly bell shaped and not heavily skewed in either direction. Because the distribution is roughly symmetric the mean is the best measure of central tendency to represent the center of this distribution. The standard deviation of 11.44 tells us that most away game scores fell within about 11 to 12 points of the average which shows the Warriors were consistent on the road. When comparing home and away performance the Warriors scored about 3 points more per game at home than away which suggests they did perform slightly better on their home court. However, the standard deviations for both home and away games are very similar at 11.69 and 11.44 which tells us the level of consistency was about the same regardless of where they played.

8. Confidence Intervals for the Average Relative Skill of All Teams in Your Team's Years

The 95% confidence interval for the average relative skill of all NBA teams during the Warriors years from 2013 to 2015 is summarized in the table below.

Table 4. Confidence Interval for Average Relative Skill of Teams in Your Team's Years

Confidence Level (%)	Confidence Interval
95%	1502.02, 1507.18

Confidence intervals are generally used to estimate the true value of a population parameter based on sample data. Instead of giving one single number as an estimate, they provide a range of values that the true population mean is likely to fall within. Based on the results, the 95% confidence interval for the average relative skill of all NBA teams from 2013 to 2015 is between 1502.02 and 1507.18. This means we are 95% confident that the true average skill level of all teams in the league during those years falls somewhere within that range. If a higher confidence level like 99% had been used the interval would have been wider to account for the extra certainty required. The probability that a team in the league has a relative skill level less than the Warrior during this period is very high which tells us the Warriors were among the more skilled teams in the league, and it would not be unusual for most other teams to fall below their skill level.

9. Confidence Intervals for the Average Relative Skill of All Teams in the Assigned Team's Years

The 95% confidence interval for the average relative skill of all NBA teams during the Bulls years from 1996 to 1998 is summarized in the table below.

Table 5. Confidence Interval for Average Relative Skill of Teams in Assigned Team's Years

Confidence Level (%)	Confidence Interval
95%	1487.66, 1493.65

The 95% confidence interval for the average relative skill of all NBA teams from 1996 to 1998 is between 1487.66 and 1493.65. This means we are 95% confident that the true average skill level of all teams in the league during those years falls somewhere within that range. If a different confidence level had been used, such as 99%, the interval would have been wider to capture that

extra level of certainty. When comparing this interval to the Warriors interval from 2013 to 2015 the average skill level of teams was higher during the Warriors years than during the Bulls years. The Warriors era interval of 1502.02 to 1507.18 is noticeably higher than the Bulls era interval of 1487.66 to 1493.65 which suggests that the overall level of competition in the NBA increased over time. This could mean that teams became more skilled overall between the late 1990s and the mid-2010s.

10. Conclusion

Overall, this project provided a thorough look at the Golden State Warriors performance from 2013 to 2015 and how they compared to the Chicago Bulls from 1996 to 1998. The data visualizations showed that the Warriors were a consistent and high scoring team with most games falling between 100 and 115 points per game. The descriptive statistics confirmed that the Warriors scored slightly more at home than away but remained consistent in both settings with similar standard deviations. The confidence intervals showed that the average skill level across the league was higher during the Warriors years compared to the Bulls years which suggests the NBA became more competitive over time. These findings are practically important for management because they show that the Warriors were performing at a high level during this period and that maintaining consistency both at home and away is key to continued success. Understanding these patterns can help management make better decisions about team strategy and areas for improvement going forward.

11. Citations

AI Usage Disclaimer:

The thoughts, concepts, and analysis presented in this report are my own. Artificial intelligence was used as a tool to assist with the explanation and interpretation of statistical findings. All final decisions regarding content and conclusions were made by the student.

- Anthropic. (2026, May 25). Project One statistical analysis assistance [Generative AI chat]. Claude. <https://claude.ai>
- FiveThirtyEight. (2019). FiveThirtyEight NBA Elo dataset. Kaggle. <https://www.kaggle.com/fivethirtyeight/fivethirtyeight-nba-elo-dataset/>